Lecture 11

Interference

Instructor: Geo Jose

Light Waves and Interference

Last time I spoke about the wave equation and various types of waves: waves in water, waves of sound, light etc. Today we will focus on light. The nature of the wave depends on the source that produces it. You are probably familiar with plane waves.

Plane Electromagnetic Waves

A plane EM wave travelling in the z direction is described by

$$\vec{E} = \vec{E}_0 \sin(kz - \omega t), \tag{1}$$
$$\vec{B} = \vec{B}_0 \sin(kz - \omega t), \tag{2}$$

where

$$k = \frac{2\pi}{\lambda}, \tag{3}$$
$$\omega = 2\pi\nu. \tag{4}$$

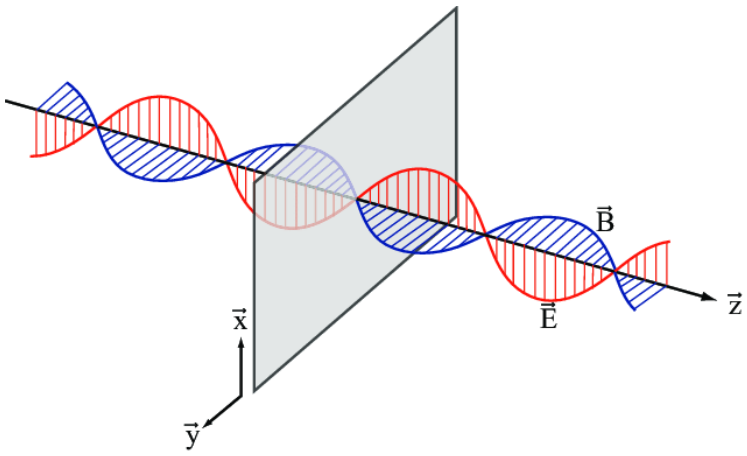


Figure 1: Plane Wave

The polarization of the EM wave is given by $\vec{E}_0$.

The plane of polarization is determined by the direction of $\vec{E}_0$ and the direction of propagation, which here is z direction.

The magnetic field part is fixed by the following stipulations:

$$\vec{B} \perp \vec{E}$$

and

$$\vec{B} \perp \hat{z}.$$

Also,

$$\vec{E} \times \vec{B}$$

points along the direction of propagation.

For example, if

$$\vec{E}_0 = E_0\hat{x},$$

then

$$\vec{B}_0 = B_0\hat{y}.$$

The three vectors $\vec{E}$, $\vec{B}$ and the direction of propagation are mutually perpendicular.

We could define a vector called Poynting vector:

$$\boxed{\vec{S} = \frac{1}{\mu_0}\vec{E} \times \vec{B}}$$

This vector is useful because it represents energy flow. Its units are

$$\text{energy} \times \frac{1}{\text{unit area} \times \text{unit time}}.$$

For the plane wave above,

$$\vec{S} = \frac{1}{\mu_0}(\vec{E} \times \vec{B}) \tag{5}$$

$$= \frac{1}{\mu_0}E_0\frac{E_0}{c}\sin^2(kz - \omega t)\,\hat{z}. \tag{6}$$

Therefore,

$$\boxed{\vec{S} = \frac{E_0^2}{\mu_0 c}\sin^2(kz - \omega t)\,\hat{z}}$$

Observe that the energy is proportional to the square of the electric field amplitude. Our perception is roughly sensitive to this.

Why is it called a plane wave?

A plane wave is called a plane wave because its wavefronts are planes. A wavefront is a set of points in space with the same phase of the wave. For the plane wave,

$$kz - \omega t = \text{constant}.$$

At a fixed time,

$$z = \text{constant}.$$

Therefore all points in the xy plane have the same phase. Hence the wavefront is a plane.

Spherical Wave

Another type of wave is a spherical wave. A spherical wave can be written schematically as

$$E(r) = A(r)\cos(kr - \omega t).$$

The wavefronts are given by

$$r = \text{constant}.$$

Therefore the wavefronts are spherical surfaces.

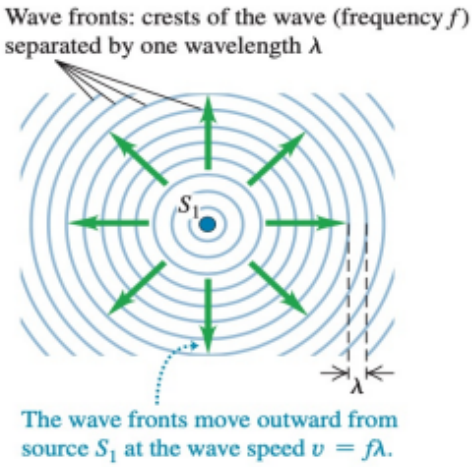


Figure 2: Spherical Waves

Interference

Very characteristic of waves is the phenomenon of interference, which is produced by two sources which produce travelling waves at the same frequency. Figure 3 is a picture showing that.

What I have drawn are spherical wavefronts .Suppose now you are at position P at a distance r_1 from source number 1 (S_1) and r_2 (S_2) from source number 2.

Then it is possible that at some observation point (like points a and b in the figure), the waves from 1 and 2 arrive in phase. A crest from 1 arrives at the same time as a crest from 2, or a valley from 1 arrives at the same time as a valley from 2. We call it **constructive interference**. It is also possible that they kill each other. This is **destructive interference**. If r_2 is $\lambda/2$ greater than r_1, waves from 2 are shifted by half a wavelength and a crest from 1 overlaps with a trough from 2, giving destructive interference.

$$r_2 - r_1 = \pm\frac{n\lambda}{2}, \qquad n = 1, 2, 3, \ldots$$

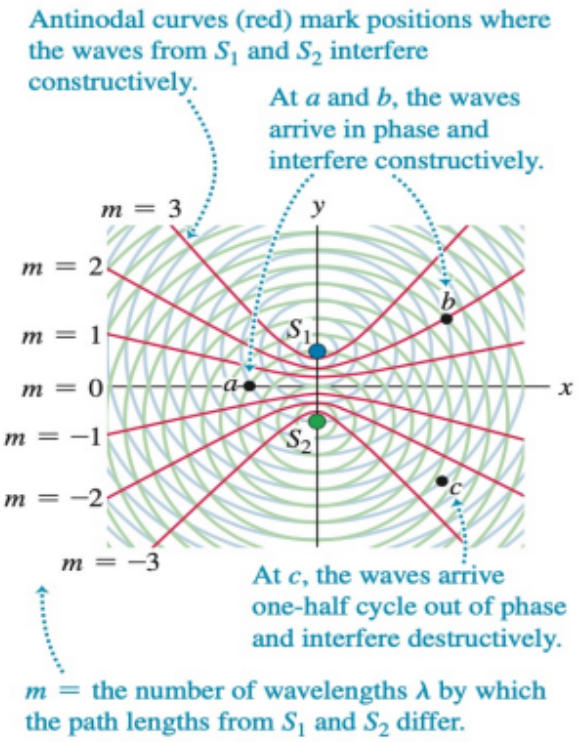


Figure 3: Interference of Spherical Waves

or, equivalently,

$$\boxed{r_2 - r_1 = \frac{(2n+1)\lambda}{2}}$$

where

$$n = 0, \pm 1, \pm 2, \pm 3, \ldots$$

for destructive interference.

Similarly, constructive interference occurs when

$$\boxed{r_2 - r_1 = n\lambda}$$

where

$$n = 0, \pm 1, \pm 2, \ldots$$

Superposition of Two Waves

The simple approach above gave us the path difference (henceforth called δ) for maxima and minima. Let us now approach it froma general point of view and obtain amplitude of the resultant wave at any given point P (which is at a distance r_1 from S_1 and r_2 from S_2) with $\delta = r_2 - r_1$. See Figure 4.

Consider two waves arriving at P. From source S_1,

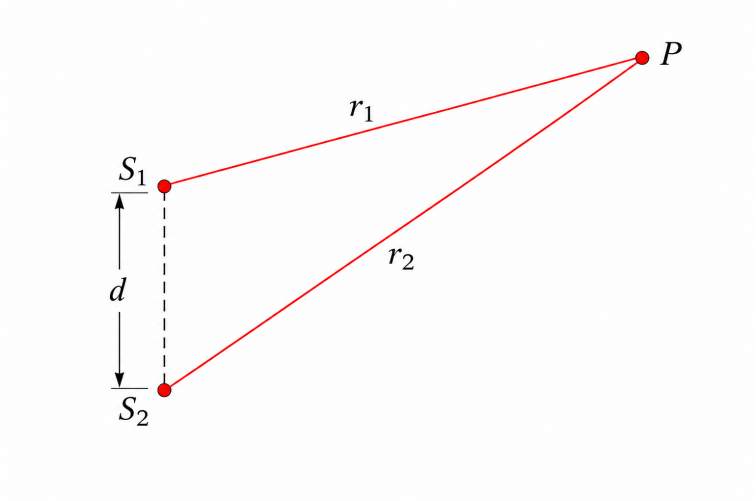


Figure 4: Set-up

$$A\cos(kr_1 - \omega t)$$

and from source S_2

$$A\cos(kr_2 - \omega t).$$

The resultant wave at P is

$$R = A\cos(kr_1 - \omega t) + A\cos(kr_2 - \omega t). \tag{7}$$

Let

$$r_2 = r_1 + \delta.$$

Then

$$R = A\cos(kr_1 - \omega t) + A\cos(kr_1 + k\delta - \omega t). \tag{8}$$

Using

$$\cos A + \cos B = 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right),$$

we obtain

$$R = 2A\cos\left(kr_1 - \omega t + \frac{k\delta}{2}\right)\cos\left(\frac{k\delta}{2}\right). \tag{9}$$

Thus,

$$\boxed{R = 2A\cos\left(\frac{k\delta}{2}\right)\cos\left(kr_1 - \omega t + \frac{k\delta}{2}\right)}$$

The amplitude is therefore

$$\boxed{A_{\text{resultant}} = 2A\cos\left(\frac{k\delta}{2}\right)}$$

Since

$$k = \frac{2\pi}{\lambda},$$

we have

$$\frac{k\delta}{2} = \frac{\pi\delta}{\lambda}.$$

Maxima and Minima

Look at the resultant amplitude $A_{\text{resultant}}$. For maxima we need,

$$\frac{k\delta}{2} = n\pi, \qquad n = 0, \pm 1, \pm 2, \ldots$$

Therefore,

$$\frac{2\pi}{\lambda}\frac{\delta}{2} = n\pi,$$

so

$$\boxed{\delta = n\lambda}$$

For minima,

$$\frac{k\delta}{2} = \frac{(2n+1)\pi}{2}, \qquad n = 0, \pm 1, \pm 2, \ldots$$

Therefore,

$$\boxed{\delta = \frac{(2n+1)\lambda}{2}}$$

These give us the saem expressions we obtained before.

Geometry of the Interference Pattern

Some geometry now.

The condition

$$r_2 - r_1 = \delta$$

defines a hyperbola. When $\delta = m\lambda$ for integer m, the hyperbola traces points where we have maxima. These are drawn in Figure 3. Take $m = 1$, that means $\delta = \lambda$, which defines a hyperbola that traces points where the first maxima occurs. Similarly, $m = 2$ defines the hyperbola that traces points where the second maxima occurs. Note: After you see the hyperbola defined below, count how many maximas a possible for a given d.

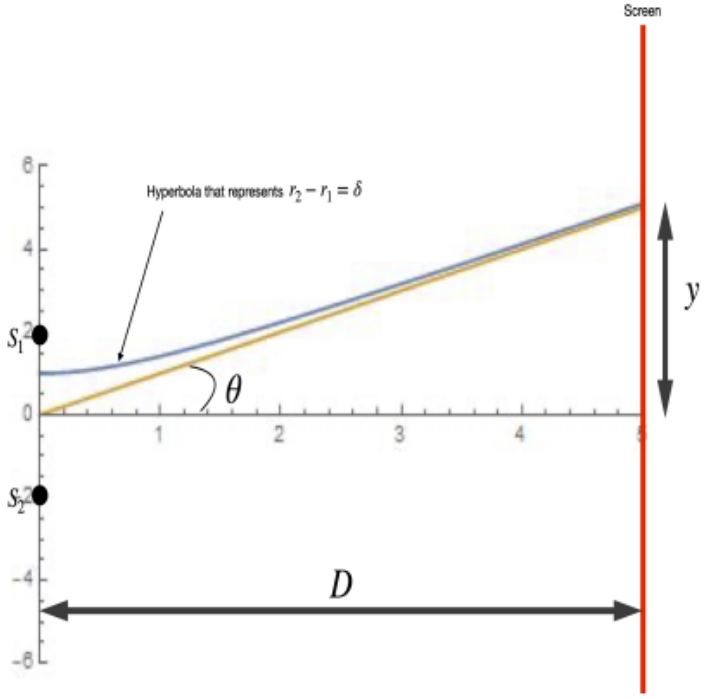


Figure 5: Hyperbola described in the text. Separation between the two sources is taken to be d

Let the separation between the two sources be d.

The hyperbola is described by

$$\boxed{\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1}$$

where

$$c^2 = a^2 + b^2.$$

For the two sources,

$$2a = \delta,$$

and

$$2c = d.$$

Therefore,

$$a = \frac{\delta}{2}, \qquad c = \frac{d}{2}.$$

Since

$$b^2 = c^2 - a^2,$$

we get

$$b^2 = \frac{d^2}{4} - \frac{\delta^2}{4} \tag{10}$$

$$= \frac{d^2 - \delta^2}{4}. \tag{11}$$

Thus,

$$\boxed{b = \frac{1}{2}\sqrt{d^2 - \delta^2}}$$

Asymptote of the Hyperbola

When is the asymptote valid?

Starting from

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1,$$

we have

$$\frac{y^2}{a^2} = 1 + \frac{x^2}{b^2}.$$

For large x,

$$x \gg b,$$

and therefore

$$\frac{x^2}{b^2} \gg 1.$$

Hence

$$\frac{y^2}{a^2} \approx \frac{x^2}{b^2}.$$

Thus,

$$\boxed{\frac{y}{x} \approx \frac{a}{b}}$$

or

$$\boxed{y = \pm \frac{a}{b} x}$$

which are the asymptotes.

For the interference geometry,

$$\frac{a}{b} = \frac{\delta}{\sqrt{d^2 - \delta^2}}.$$

Therefore,

$$\boxed{\frac{y}{x} = \frac{\delta}{\sqrt{d^2 - \delta^2}}}$$

for the asymptotic region. But remember that y/x is $tan\theta$ where θ is the angle made by the asymptote with the x-axis.

Then

$$\tan\theta = \frac{y}{x}.$$

For the hyperbola,

$$\tan\theta = \frac{a}{b}.$$

Using

$$a = \frac{\delta}{2}, \qquad b = \frac{1}{2}\sqrt{d^2 - \delta^2},$$

we obtain

$$\boxed{\tan\theta = \frac{\delta}{\sqrt{d^2 - \delta^2}}}$$

Now,

$$\cot^2\theta = \frac{d^2 - \delta^2}{\delta^2} = \frac{d^2}{\delta^2} - 1.$$

Therefore,

$$1 + \cot^2\theta = \frac{d^2}{\delta^2}.$$

Since

$$1 + \cot^2\theta = \csc^2\theta,$$

we get

$$\csc^2\theta = \frac{d^2}{\delta^2}.$$

Hence,

$$\csc\theta = \frac{d}{\delta},$$

and therefore

$$\boxed{\sin\theta = \frac{\delta}{d}}$$

or

$$\boxed{d\sin\theta = \delta}$$

Thus we now have one more capability, we can predict the angle from the centerline θ where δ path difference occurs. See Figure 5 and make sure you understand the angle θ that the observation point makes with the x axis. In particular if we plug in the values of δ corresponding to maxima and minima, we can predict the angles at which maxima and minima occur.

Angular Positions on the Screen

From

$$d\sin\theta = \delta,$$

for constructive interference,

$$\delta = n\lambda,$$

so

$$\boxed{\sin\theta = \frac{n\lambda}{d}}$$

For destructive interference,

$$\delta = \frac{(2n+1)\lambda}{2},$$

so

$$\boxed{\sin\theta = \frac{(2n+1)\lambda}{2d}}$$

Fringe Positions on the Screen

Now let us say we want to be able to predict the distance along y on the screen where these maxima and minima occur. See y marked in Figure 5. For that we need an approximation:

For lower-order fringes,

$$\sin\theta \approx \tan\theta.$$

Also, from the geometry of the screen,

$$\tan\theta = \frac{y}{D},$$

where D is the distance from the sources to the screen.

Therefore,

$$\sin\theta \approx \frac{y}{D}.$$

Using

$$d\sin\theta = \delta,$$

we get

$$d\frac{y}{D} = \delta.$$

Hence,

$$\boxed{y = \frac{\delta D}{d}}$$

This gives the linear distance of a fringe on the screen.

For constructive interference,

$$\delta = n\lambda.$$

Therefore,

$$y_n = \frac{n\lambda D}{d}.$$

Thus,

$$\boxed{y_n = \frac{n\lambda D}{d}}$$

For destructive interference,

$$\delta = \frac{(2n+1)\lambda}{2}.$$

Therefore,

$$y_n = \frac{(2n+1)\lambda}{2}\frac{D}{d}.$$

Thus,

$$\boxed{y_n = \frac{(2n+1)\lambda D}{2d}}$$

Validity of the Asymptote and the Small-Angle Approximation

The condition we used above for using the asymptote is

$$\boxed{x \gg b}$$

or

$$\boxed{x \gg \frac{1}{2}\sqrt{d^2 - \delta^2}}$$

Observe that r_1, r_2, d form three sides of a triangle. Therefore $r_2 \leq r_1 + d$. Which means $r_2 - r_1 = \delta \leq d$. The largest value of $\sqrt{d^2 - \delta^2}$ is in fact d. Therefore we can use the asymptotic formula if we keep our screen far away from the sources ($x >> d$). Now let us look at the small angle approximation.

For the hyperbola,

$$tan\theta = \frac{y}{x} = \frac{a}{b} = \frac{\delta}{\sqrt{d^2 - \delta^2}}.$$

Also,

$$\sin\theta = \frac{\delta}{d}.$$

The approximation

$$\sin\theta \approx \tan\theta$$

requires a small angle.

Since

$$\sqrt{d^2 - \delta^2} \approx d$$

when

$$\boxed{d \gg \delta,}$$

we have

$$\tan\theta = \frac{\delta}{\sqrt{d^2 - \delta^2}} \approx \frac{\delta}{d} = \sin\theta.$$

Thus the small-angle approximation is valid when

$$\boxed{d \gg \delta.}$$

What does this mean? If your observation point P is on the x axis, $\delta = 0$. If P is on the y axis, $\delta = d$. As you move you observation point up from x axis, δ keeps increasing. Therfore to obey $d \gg \delta$, you have to stay close to x axis, which in other words is small angles θ.